MIL-F-37576
4 October 1976

MILITARY SPECIFICATION

FRAME, OPHTHALMOLOGICAL, TRIAL LENS SET

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of revolving 3 cell ophthalmological trial lens frame, with corneal sighting device.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-392	- Plastic Molding Material, Acetal, Injection and Extrusion.
L-P-393	- Plastic Molding Material, Polycarbonate, Injection and Extrusion.
L-P-401	- Plastic Molding Material, Urea-Formaldehyde.
QQ-A-591	- Aluminum Alloy Die Castings.
PPP-B-585	- Box, Wood, Wirebound.
PPP-B-601	- Box, Wood, Cleated-Plywood.
PPP-B-621	- Box, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC-6515

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MIL-P-116 - Preservation, Method of.
MIL-L-10547 - Liners, Case and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.
MIL-S-46889(MI) - Steel, Spring, Chromium Nickel Alloy.

STANDARDS

FEDERAL

FED-STD. No. 151 - Metal, Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedure and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 221 Aluminum Alloy Extruded Bars, Rods, Wire, Shapes and Tubes

(Application for copies should be addressed to American Society for Testing and Materials, 1903 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. Materials used for the frame shall be as specified herein, or equal, and suitable for the intended purpose.

3.1.1 Temple bar, beam and spring, and cell and nosepiece extension spring. The temple bar, beam and spring and the cell and nosepiece spring shall be spring steel, high carbon AISI 1095 or chromium-nickel alloy in accordance with MIL-S-46889 (MI).

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3.1.2 Eyepiece, head and hinge. The eyepiece (loupe) head and hinge shall be fabricated of die cast aluminum conforming to QQ-A-591.

3.1.3 Knob, knob screw, nosepiece extension and housing. All knobs, knob screws and the nosepiece extension tube and housing shall be aluminum extrusion in accordance with ASTM B 221.

3.1.4 Cylinder cell and gear. Cylinder cells and gears shall be fabricated of molded acetal plastic conforming to L-P-392. Color shall be black.

3.1.5 Earpiece. The temple earpiece shall be fabricated of polycarbonate plastic conforming to L-P-393. Color shall be white or flesh colored.

3.1.6 Noserest. The noserest for the bridge shall be molded from urea-formaldehyde conforming to L-P-401. Color shall be white or flesh colored.

3.1.7 Scales. The pupillary distance (PD) and cylinder axis scales shall be clear anodized aluminum sheet, satin finish.

3.1.8 Hardware screws. Miscellaneous screws shall be aluminum or stainless steel.

3.2 Style and design. The frame shall be an ophthalmolgical trial frame for use with a trial lens set in performing ocular refractions. The frame shall be so designed and constructed as to be adjustable for a universal fit of all heads and bridges without discomfort to the patient. The trial lens frame shall be a revolving three cell unit with a corneal sighting device, 48 mm to 80 mm PD adjustment, and extensive range of horizontal and vertical bridge movement. The bridge shall be provided with a noserest for comfortable fit. Other features shall include leaf spring temples, adjustable temples with formed plastic paddle tips and gear and pinion operated cylinder cell rotation. The trial frame shall conform to the style and design shown in figure 1. It shall be capable of accomodating four 1-1/2 inch trial lens rings in one single stationary cell and a triple rotating cell. Cylinder axis and PD scales shall be clear satin finish with black numbers on the light background, clearly applied for legibility. Pupillary distance drive shafts and cylinder cell rotating mechanisms shall be appropirately housed to reduce contamination by body oils and foreign particles. The temples shall be adjustable from 5 degrees above to 15 degrees below center and from approximately 3-1/2 to 5 inches in length. The temple tips shall hold the frame snugly against the patients's head.

3.2.1 Dimensions. The distance from loupe end piece to loupe end piece shall be 5-1/4 ± 1/16 inch, and from botton of loupe to top of head shall be 2-3/8 ± 1/16 inch. Diameter of eye opening shall be 31-1/2 ± 1/2 mm.

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3.3 Finish. The loupes, head, temple hinge and sighting device shall be anodized and black enameled. All other aluminum parts shall be machined smooth and clear anodized. The temples shall be bright nickel-chromium plated.

3.4 Service data. The contractor shall enclose, with each unit, two copies each of one or more booklets which contain complete instructions for installation, operation and maintenance, and a list of component parts. Each part of the unit other than common fasteners, shall be identified in the service data and parts list by the manufacturer's part number. In addition, the contractor shall mail 15 copies of the service data to the contracting officer for distribution. Should the contractor have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of the 15 additional copies shall be at the option of the Government.

3.5 Identification marking. The frame shall be clearly and legibly marked with the manufacturer's name or registered trademark.

3.6 Workmanship. Workmanship shall be first class throughout. Frames shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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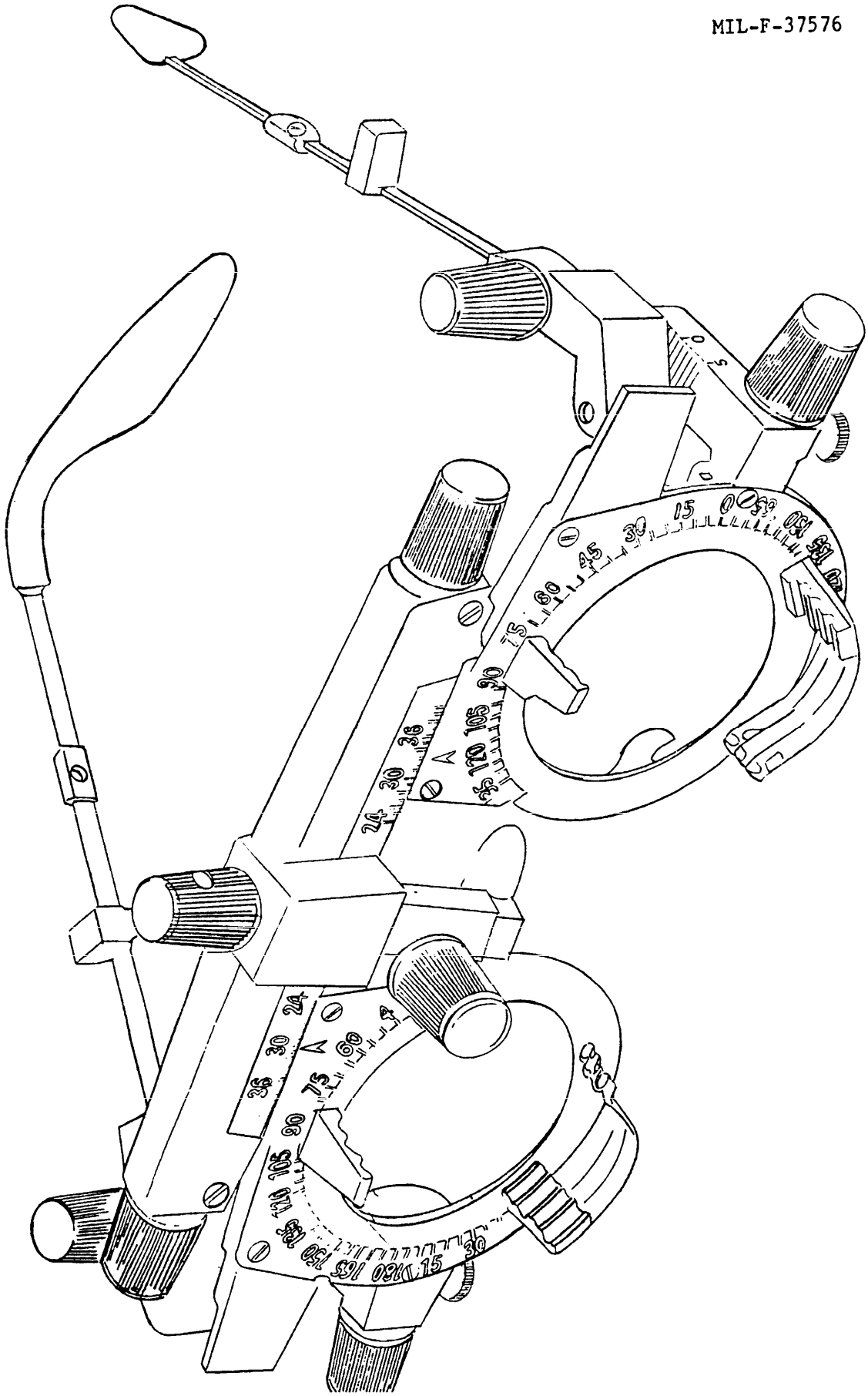


FIGURE 1. Trial frame

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the metal or component, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demostrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer of supplier, the National Stock Number (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table I. The unit of product for examination purposes shall be one frame.

TABLE I. Sampling for examination.

Type of defect	Inspection level	AQL (percent defective
Major	II	1.0
Minor	II	2.5
Dimension	S-2	1.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. Where feasible the same sample shall be used for the determination of two or more test characteristics. The unit of product for test purposes shall be one frame.

TABLE II. Sampling for tests.

Characteristic	Requirement	Inspection level	AQL (Percent defective)
Metals composition	Section 3	2/	1/
Plastic tests	3.1.4, 3.1.5 & 3.1.6	3/	1/

1/ Acceptance number shall be zero.
2/ Four ounces of each lot of metal used for the frame.
3/ See basic specification.

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4.3 Examination. Examination of the frame shall be conducted to determine specification compliance. Examination for defects shall be conducted in accordance with but not restricted to those listed in table III. Additional defects shall be classified in accordance with MIL-STD-105.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Component missing, broken or cracked.
102	Component design not as specified.
103	Temple not adjustable as required.
104	Incorrect range or incorrect increments of PD scale.
105	Incorrect range or incorrect increments of cylinder scale.
106	Nose rest not adjustable.
107	Gear and pinion will not operate the cells.
108	Not assembled as specified or poorly accomplished.
Minor	
201	Metal parts not free of sharp edges and burrs.
202	Temple not adjustable from 3-1/2 to 5 inches.
203	Loose screws or improper size.
204	Black enamel finish missing.
205	Anodizing or plating missing.
206	Wrong color temple tip or nose rest.

4.3.1 Dimensional examination. Examination shall be made to determine compliance with the applicable dimensional requirements. Any dimension that is not within the specified tolerance shall be classified as a defect.

4.3.2 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A quality conformance inspection requirements of MIL-P-116. The inspection of the package and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Subsidiary specifications. The metal and plastic material shall be tested for conformance with the chemical composition and physical properties of the referenced subsidiary specifications. Where samples and test methods are not defined herein the basic specification shall apply.

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4.4.2 Chemical analysis of the metals. The samples selected in accordance with table II and 4.4.1 shall be analyzed in accordance with methods 111.2 or 112.2 of Federal Test Method Standard No. 151 for compliance with requirements of section 3. In case of dispute, method 111.2 shall be used for referee.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning, drying and preservative. Each frame shall be cleaned and dried in accordance with the applicable process specified in MIL-P-116. Preservative shall not be applied.

5.1.1.2 Unit package. Each frame shall be inserted into a paper envelope or wrapped in paper barrier material and further packaged into a paper board box of appropriate size. Closure of box shall be adequate to prevent accidental opening. Packaging shall be method III, MIL-P-116.

5.1.2 Level C. Frames shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Forty eight (48) frames shall be packed in an exterior container designed for a type 2 load constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. Forty eight (48) frames shall be packed in an exterior container designed for a type 2 load constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

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5.2.3 Level C. The packaged frames shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Intended use. The trial frame is intended for professional use with trial lenses in testing visual acuity.

6.2 Ordering data. Purchasers should exercise the preferred options offered herein, and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Applicable levels of packaging and packing (see 5.1 and 5.2).
(c) National stock number.

6.3 NSN coverage. This specification covers the following item appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00-346-2240	FRAME, OPHTHALMOLOGICAL TRIAL LENS SET, with Corneal Sighting Device

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DSA-DM

Project No. 6515-2029

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-F-37576 FRAME, OPHTHALMOLOGICAL, TRIAL LENS SET

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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